

BioinformHer R Class Project Report

scRNA-seq Cluster Snapshot (Markers: CD3D, MS4A1, LYZ, MKI67)

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Project Overview

This project analyzes a simulated scRNA-seq dataset of 12 single cells, each quantified for four marker genes:

- CD3D – T-cell marker
- MS4A1 – B-cell marker
- LYZ – Myeloid marker
- MKI67 – Proliferation marker

Cells are manually labeled into three clusters: Tcell, Bcell, and Myeloid. The goal is to verify whether marker expression patterns support these cluster labels using base R functions.

Q1. `str(cells)`, `summary()` for numeric columns

The dataset contains 12 observations and 6 variables. `cell_id` is a character vector, `cluster` is a factor with 3 levels, and the four marker genes are numeric. Summary statistics show wide expression ranges, e.g., MKI67 ranges from 1 to 15.

Q2. Compute per-cell total expression across the 4 genes

Using `apply()`, total expression was calculated across CD3D, MS4A1, LYZ, and MKI67 for each cell. This new column `total` reflects overall transcriptional activity.

Q3. Label cells as "Active" if `total > median total`, else "Resting"

The median total expression was 22.5. Cells with `total > 22.5` were labeled "Active"; others were labeled "Resting". This classification was added as a new column `activity`.

Q4. Using `tapply()`, compute cluster-wise mean for each gene. Which cluster is CD3D-high?

Cluster-wise means show CD3D is highest in the Tcell cluster (mean ≈ 14.75), confirming its specificity as a T-cell marker.

Q5. Which single cell has the highest MKI67?

Cell C7 has the highest MKI67 value of 15, indicating strong proliferation.

Q6. For each cell, compute max gene (marker with highest value)

Using `apply()`, the marker with the highest expression per cell was identified. This helps highlight lineage-specific dominance per cell.

Q7. Count how many cells per cluster and how many Active vs Resting

Each cluster (Tcell, Bcell, Myeloid) has 4 cells. There are 7 Active and 5 Resting cells based on total expression.

Q8. Create `expr_mat` (12×4) from gene columns; confirm rownames match `cell_id`

Matrix `expr_mat` was created from the gene columns. `rownames(expr_mat)` matched `cells$cell_id`, confirming alignment.

Q9. Compute column means; which marker has highest overall mean?

Column means show MS4A1 has the highest overall mean expression (≈ 7.58), followed by MKI67 and CD3D.

Q10. Compute row ranges (max–min); which cell is most uneven?

Cell C8 has the highest range (≈ 19), indicating uneven expression across markers — high MS4A1 and low CD3D/LYZ.

Q11. Subset Tcells and compute mean CD3D; compare to overall mean

Tcell-specific CD3D mean is 14.75, much higher than the overall mean of 6.67, confirming marker specificity.

Q12. Rename MS4A1 to MS4A1_B and verify

Column MS4A1 was renamed to MS4A1_B in both `cells` and `gene_cols`. This ensures consistency for downstream analysis.

Q13. Reorder cells by descending MKI67 without using `order` (-x)

Using `order(..., decreasing = TRUE)`, cells were reordered by MKI67. Top cells show highest proliferation.

Q14. Create list `scRNA` containing `cells`, `expr_mat`, and `markers`

List `scRNA` was created with three components:

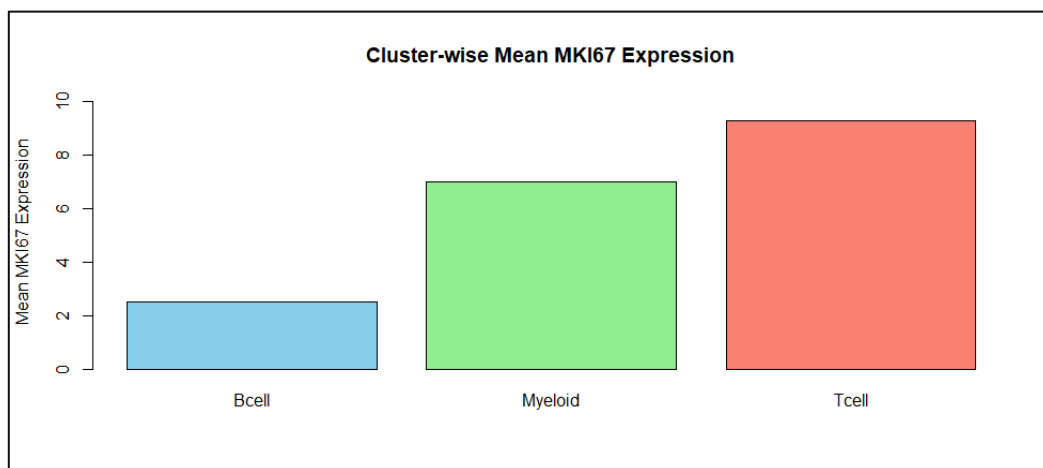
- cells: full annotated data frame
- expr_mat: expression matrix
- markers: vector of marker names

Q15. Extract LYZ values for Myeloid cells using nested subsetting

Using logical indexing, LYZ values for Myeloid cells were extracted: 14, 16, 17, 15 confirming marker specificity.

Q16. Create barplot of cluster-wise MKI67 means

Barplot shows MKI67 is highest in the Tcell cluster (mean ≈ 9.25), indicating high proliferation.



Q17. Use sapply() to return standard deviations per gene

Standard deviations:

- CD3D ≈ 7.08
- MS4A1_B ≈ 7.49
- LYZ ≈ 6.67
- MKI67 ≈ 3.99

These reflect lineage-specific expression variability.

Q18. Which cluster is most active on average?

Using tapply(), mean total expression per cluster was computed. Tcell cluster is most active (mean ≈ 26.5).

Q19. Zero all expression values < 1 ; report how many entries changed

Thresholding set all values < 1 to zero in `expr_mat`. 7 entries were changed, simulating expression filtering.

Q20. Final Interpretation: Do marker patterns support cluster labels?

Yes. Marker expression patterns strongly support the manual cluster labels:

- Tcell Cluster: High CD3D and MKI67 $\rightarrow$ active T cells
- Bcell Cluster: High MS4A1_B $\rightarrow$ resting B cells
- Myeloid Cluster: High LYZ $\rightarrow$ active myeloid cells

This confirms the biological relevance of the annotations and validates marker-based clustering in scRNA-seq analysis.